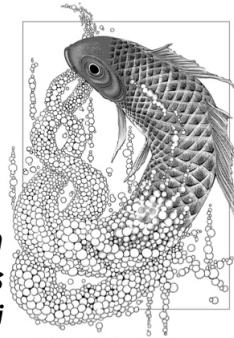


# ***The origin of ferromagnetism and the Hubbard model***

***appendix 1: Perron-Frobenius theorem for  
real symmetric matrices***

***Advanced Topics in  
Statistical Physics  
by Hal Tasaki***



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# notation

$N$  dimensional linear space

$$\text{vector } \psi = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix} = (\psi_i)_{i=1, \dots, N} \quad (1)$$

$$\psi_i \in \mathbb{C} \quad (2)$$

$$\text{we also write } (\psi)_i = \psi_i \quad (3)$$

$$\text{inner product } \langle \psi, \psi \rangle := \sum_{i=1}^N \psi_i^* \psi_i \quad (4)$$

$$\text{norm } |\psi| = \sqrt{\langle \psi, \psi \rangle} \quad (5)$$

$$\text{matrix } M = \begin{pmatrix} m_{11} & \dots & m_{1N} \\ \vdots & \ddots & \vdots \\ m_{N1} & \dots & m_{NN} \end{pmatrix} = (m_{ij})_{i,j=1, \dots, N} \quad (6)$$

$$m_{ij} \in \mathbb{C} \quad (7)$$

$$\text{we also write } (M)_{ij} = m_{ij} \quad (8)$$

## Theorem

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let  $M = (M_{ij})_{i,j=1,\dots,N}$  be  $N \times N$  real symmetric matrix, i.e.  $M_{ij} = M_{ji} \in \mathbb{R}$ , and further assume that

- (i)  $M_{ij} \leq 0$  for any  $i \neq j$   $i$  and  $j$  are "connected" by nonzero entries
- (ii) for any  $i, j$  with  $i \neq j$ , there is a sequence  $i_0, i_1, \dots, i_k$  s.t.  
 $i_0 = i$ ,  $i_k = j$ , and  $M_{i_k i_{k+1}} \neq 0$  for all  $k=1, \dots, k-1$ , where  $k \leq N$

then the lowest eigenvalue  $\mu_0$  of  $M$  is nondegenerate and the corresponding eigenvector  $\psi = (\psi_i)_{i=1,\dots,N}$  can be chosen so that  $\psi_i > 0$  for all  $i=1, \dots, N$ .

(The theorem is valid for non-symmetric matrices but the proof is harder)

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lemma 1

let  $\psi = (\psi_i)_{i=1, \dots, D}$  be an eigenvector of  $M$  ( $M\psi = \lambda\psi$ ,  $|\psi| \neq 0$ ) such that  $\psi_i \geq 0$  for all  $i$ . then  $\psi_i > 0$  for all  $i=1, \dots, D$

proof take  $\alpha > 0$  such that  $1 - \alpha\lambda > 0$  and  $1 - \alpha m_{ii} > 0$  for all  $i$ .

define  $\tilde{M} = I - \alpha M$ .

then  $(\tilde{M})_{ij} \geq 0$  for all  $i, j$ ,  $(\tilde{M})_{ii} > 0$  for all  $i$ , and

$$(\tilde{M}^N)_{ij} = \sum_{i_1, i_2, \dots, i_{N-1}=1}^N (\tilde{M})_{ii_1} (\tilde{M})_{i_1 i_2} \dots (\tilde{M})_{i_{N-1} j} > 0 \text{ for any } i, j \quad (1)$$

$\geq 0$  in general

$> 0$  for at least one sequence because of (ii)

now  $\tilde{M}^N \psi = (1 - \alpha\lambda)^N \psi$ , i.e.  $\sum_j (\tilde{M}^N)_{ij} \psi_j = (1 - \alpha\lambda)^N \psi_i \quad (2)$

$$\psi_i = \underbrace{(1 - \alpha\lambda)^{-N}}_{> 0} \sum_j \underbrace{(\tilde{M}^N)_{ij}}_{> 0} \underbrace{\psi_j}_{> 0 \text{ for at least one } j} > 0 \quad (3) \text{ for all } i //$$

## lemma 2

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it suffices to consider only eigenvectors with real components.

$$M\psi = \lambda\psi$$

$$(\psi)_i \in \mathbb{R} \text{ for all } i$$

proof

$$\psi = (\psi_i)_{i=1, \dots, N}, \quad \psi^* = (\psi_i^*)_{i=1, \dots, N} \quad (1)$$

$$M\psi = \lambda\psi \iff M\psi^* = \lambda\psi^* \quad (2)$$

$$\psi_R = \frac{1}{2}(\psi + \psi^*), \quad \psi_I = \frac{1}{2i}(\psi - \psi^*) \quad (3)$$

at least one of  $\psi_R, \psi_I$  is nonzero and is an eigenvector.

lemma 3 (variational principle) $\mu_0$ : the lowest eigenvalue of  $M$ 

$$\langle \psi, (M - \mu_0) \psi \rangle \geq 0 \text{ for any vector } \psi \quad (1)$$

$$\langle \psi, (M - \mu_0) \psi \rangle = 0 \iff M\psi = \mu_0 \psi \quad (2)$$

proof easyorthonormal basis  $\{\psi_j\}_{j=0,1,\dots,N-1}$  with  $M\psi_j = \mu_j \psi_j$  (3) $\mu_j \geq \mu_0$  for all  $j$ 

$$\psi = \sum_{j=1}^N \alpha_j \psi_j \quad (4)$$

$$\langle \psi, (M - \mu_0) \psi \rangle = \sum_{j=1}^N (\mu_j - \mu_0) |\alpha_j|^2 \quad (5) \rightarrow (1), (2) \text{ follow} //$$

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# proof of theorem

let  $\psi = (\psi_i)_{i=1, \dots, N}$  be s.t.  $M\psi = \mu_0 \psi$ ,  $\psi_i \in \mathbb{R}$  for all  $i$ , and  $|\psi| \neq 0$

We may assume  $\psi_j > 0$  for some  $j$

lemma 2

let  $I_+ = \{i \mid \psi_i > 0\} \neq \emptyset$ ,  $I_- = \{i \mid \psi_i < 0\}$  (1)

for  $i \in I_+$ ,  $j \in I_-$  or  $i \in I_-$ ,  $j \in I_+$ ,  $\psi_i m_{ij} \psi_j$   $\geq 0$  (2)  
 $\leq 0$

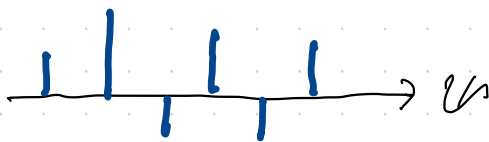
$$\langle \psi, M\psi \rangle = \sum_{i,j=1}^N \psi_i m_{ij} \psi_j \geq \sum_{i,j \in I_+} \psi_i m_{ij} \psi_j + \sum_{i,j \in I_-} \psi_i m_{ij} \psi_j$$

$$= \langle \psi_+, M\psi_+ \rangle + \langle \psi_-, M\psi_- \rangle \quad (3)$$

where  $(\psi_+)_{i} = \max\{\psi_i, 0\}$  (4)

$(\psi_-)_{i} = \min\{\psi_i, 0\}$  (5)

$$\langle \psi, \psi \rangle = \langle \psi_+, \psi_+ \rangle + \langle \psi_-, \psi_- \rangle \quad (6)$$



$$\langle \psi, M \psi \rangle \geq \langle \psi_+, M \psi_+ \rangle + \langle \psi_-, M \psi_- \rangle \quad (1)$$

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$$\langle \psi, \psi \rangle = \langle \psi_+, \psi_+ \rangle + \langle \psi_-, \psi_- \rangle \quad (2)$$

$$0 = \langle \psi, (M - M_0) \psi \rangle \geq \underbrace{\langle \psi_+, (M - M_0) \psi_+ \rangle}_{\text{lemma 3} \rightarrow \geq 0} + \underbrace{\langle \psi_-, (M - M_0) \psi_- \rangle}_{\rightarrow \geq 0} \quad (3)$$

$$\text{therefore } \langle \psi_+, (M - M_0) \psi_+ \rangle = 0 \quad (4)$$

$\downarrow$  lemma 3

$$M \psi_+ = M_0 \psi_+ \quad (5)$$

$$(\psi_+)_i \geq 0 \text{ for all } i \text{ and } |\psi_+| \neq 0 \xrightarrow{\text{lemma 1}} (\psi_+)_i > 0 \text{ for all } i$$

$$\underline{\psi_i > 0 \text{ for all } i}$$

## ▷ nondegeneracy

suppose there are  $\psi_1, \psi_2$  s.t.

$$M\psi_1 = \mu_0 \psi_1, \quad M\psi_2 = \mu_0 \psi_2 \quad (1)$$

We can assume  $(\psi_1)_i > 0, (\psi_2)_i > 0$  for all  $i$   
fix  $j_0$  and let

$$\psi = \psi_1 - \frac{(\psi_1)_{j_0}}{(\psi_2)_{j_0}} \psi_2 \quad (2)$$

Then  $M\psi = \mu_0 \psi$ ,  $\psi_i \in \mathbb{R}$  for all  $i$  and  $\psi_{j_0} = 0$

it is only possible that  $|\psi| = 0$ , i.e.,  $\psi_1 \propto \psi_2$